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æ:// — The Conductor Model

Sovereign Authority, Digital Twins, and the Right to Reflect

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æ:// · sovereign private client mesh

ABSTRACT

Human–agent systems today are *capacitive*: they scale the number of agents one person can supervise, but not the *authority* those agents may exercise. We argue that the missing layer is not infrastructure but the product itself — a *conductor*: an intermediate process through which one human directs many agents, and through which N receipts are consolidated into one verified truth. We specify the conductor's five duties (translate, arbitrate, fan out, fan in, hold memory) and identify *fan-in* — consolidating N receipts into a single verified truth — as the unsolved gap. We then extend the model vertically into a four-layer stack: *sovereign* (cryptographic custody and authority), *private client* (local-first surface), *digital twin* (conductor memory), and *digital human* (voice and face). We observe that recent full-duplex speech models such as NVIDIA PersonaPlex, and digital-human pipelines such as NVIDIA ACE/Tokkio, solve the *projection* layer — voice, face, and natural turn-taking — while providing no cryptographic identity. We show that this missing layer is exactly what a sovereign signing substrate supplies, and we propose treating every full-duplex conversational act as a *signed dispatch* that passes the same gate as any other action. We give a minimal specification of the receipt rail and report a working instance (5 waves, 20 tasks, 20/20 verified). Our central claim: NVIDIA provides the mirror; the sovereign layer owns the thing mirrored and the right to reflect it.

1. Introduction

A one-person company is not a company with one employee. It is a company whose *authority* does not scale with headcount. The founder cannot hire, cannot delegate custody, and cannot personally verify the work of twenty agents at once. Yet the tools available — LLM APIs, agent frameworks, orchestration platforms — are all *capacitive*: they increase throughput without increasing authority.

We contend that the layer between the human and the fleet is not plumbing. It is the product. We call it æ://, and this paper specifies it.

Two claims organise the paper. First, that the correct abstraction is a *conductor*, and that the conductor's hard problem is not dispatch (fan-out) but *consolidation* (fan-in): turning N agent receipts into one verified truth. Second, that the same authority substrate that makes fan-in trustworthy is what a *digital human* needs and does not have — and that a full-duplex conversational model is, structurally, a stream of *actions* that must pass the same gate.

2. The Conductor Model

2.1 Definition

Let H be a single human principal and A = {a₁, …, a_N} a set of N autonomous agents. A conductor C is an intermediate process such that

```
H → C    (intent)
C → aᵢ   (dispatch)
aᵢ → C   (receipt)
C → H    (consolidated truth)
```

C is the only process that both issues and receives. The human never addresses an agent directly; an agent never reports to the human directly. Identity is *relational*: an agent is defined by its position in this graph, not by an intrinsic property.

2.2 The Five Duties

1. **Translate.** Convert natural-language intent into typed dispatches: a capability address, a task specification, and an authority scope.
2. **Arbitrate.** Every dispatch must be authorised by a capability. Custody of the signing key remains with H. The conductor may *propose*; it may not *approve*.
3. **Fan out.** Partition work into *waves* — batches dispatched in parallel across the fleet.
4. **Fan in.** Collect receipts from each agent and consolidate them into a single verified truth. *This is the critical gap.*
5. **Hold memory.** Maintain a digest — a self-referencing record of what was dispatched, completed, and outstanding — across instances.

2.3 The Gap: Fan-In

Fan-out is largely solved: partition, dispatch, parallelise. Fan-in is not. The consolidation of N receipts into one truth requires

- **Addressability**: each receipt cites the capability address that produced it;
- **Verifiability**: each receipt carries a signature checkable against that capability's public key;
- **Deduplication**: where several agents attempt the same task, select one winner;
- **Consolidation**: reduce to a single boolean, PASS or FAIL.

We call this structure the **receipt rail**. Without it, the conductor is a load balancer, not a conductor.

3. The Vertical Stack

The conductor sits atop a four-layer stack:

```
DIGITAL HUMAN   (voice + face) — the mirror
DIGITAL TWIN    (conductor memory) — the thing mirrored
```

```
PRIVATE CLIENT   (local-first surface) — where the twin lives
      SOVEREIGN   (custody + authority) — who may act
```

Each layer depends on the one below: the human cannot speak without the twin's memory; the twin cannot remember without the client's storage; the client cannot store without the sovereign's custody.

3.1 Sovereign

The sovereign layer holds an Ed25519 signing key and is the only layer that can authorise an action. The human is its sole custodian. The conductor may request a signature; the sovereign may refuse.

3.2 Private Client

The private client is the local-first interface — the human's name, logo, and configuration, and the place the twin resides when not projected. It is personal in the legal sense: authority and credentials are never exported.

3.3 Digital Twin

The twin is the conductor's persistent memory, *feed-informed*: it updates as receipts arrive and supplies the digital human's responses. It is the record of what the human intended and what was done.

3.4 Digital Human

The digital human is the projection. Recent full-duplex speech models make this layer newly tractable: NVIDIA *PersonaPlex* [1] is a 7B full-duplex speech-to-speech model, built on *Moshi* [2], that allows voice conditioning and role control through a text prompt, and that handles interruptions, backchannels, and turn-taking with reported latencies near 0.17 s (turn-taking) and 0.24 s (interruption). NVIDIA *ACE*/Tokkio [3,4] supplies the 3D animation pipeline (Audio2Face). Together these provide voice, face, and presence. They do not provide cryptographic identity.

4. The Gate: Full-Duplex as Signed Acts

4.1 The Problem

A full-duplex model listens and speaks simultaneously, interrupting, yielding, and backchannelling in real time. Each such behaviour is an *act* — a change to the shared state of a conversation. In a single-agent system an act is cheap. In a conductor system, an act by the digital human is an act by *all agents the human represents*, and must therefore pass the gate.

4.2 The Proposal

We propose that every full-duplex act be treated as a dispatch:

1. the twin generates a candidate act (speak, pause, yield, interrupt);
2. the conductor validates it against the human's current intent and scope;
3. the sovereign signs the act as a `sign_card` capability;
4. the act is projected through the digital human;
5. a receipt is written to the rail.

This is not throttling; it is *authority enforcement*. The digital human may speak freely — but only for the conductor, and only within the conductor's scope.

4.3 The Canonical Separation

> **NVIDIA provides the mirror; æ:// owns the thing mirrored and the right to reflect it.**

The mirror is a solved problem. The authority to reflect — to decide whose face may speak, and what it may say — is not.

5. The Receipt Rail: A Minimal Specification

The rail is a single JSON document, versioned by schema:

```
{
  "schema": "receipt-rail/v1",
  "batches": [
    {"wave": "daollc-content", "status": "complete",
     "tasks": 5, "complete": 5,
     "verb_list": ["scan","extract","claim","complete","verify"]}
  ],
  "total_waves": 5, "total_tasks": 20, "total_complete": 20,
  "consolidated_verdict": "PASS - 20/20 verified via fleet waves",
  "topology": "one human -> waves (fan-out) -> agents -> receipts (fan-in)"
}
```

Each batch is a *wave*; each wave is a fleet with receipt tasks. A working instance is deployed and verifies 20/20 tasks across 5 waves. The capability that authorises a dispatch binds the verb, task identifier, subject, expiry, and a single-use nonce, and is signed with Ed25519.

6. Related Work

Full-duplex speech. PersonaPlex [1] extends Moshi [2] with voice and role control; both replace the cascaded ASR→LLM→TTS pipeline with a single dual-stream model. We adopt this class as the projection layer and observe that it has no notion of authority.

Digital humans. NVIDIA ACE [3] and the Tokkio reference workflow [4] compose NIM microservices (Riva ASR/TTS, Audio2Face, an LLM) into a lifelike avatar. The blueprint is enterprise-facing and identity-agnostic.

Mixture of agents. MoA [5] improves generation by combining models. We extend the idea from generation to *execution*: dispatching and verifying actions, with a signed receipt as the unit of trust.

7. Limitations

- **Hardware.** The reference environment (NVIDIA RTX 3050, 6 GB, cc 8.6) cannot hold PersonaPlex-7B in bf16 (≈14 GB). A q4_k quantisation (≈4.5 GB) is the only plausible local path and has not been measured; the ACE/Tokkio blueprint targets datacenter GPUs.
- **Gate latency.** Signing every conversational act adds latency whose effect on naturalness we have not measured.
- **Cross-instance fan-in.** The reported rail consolidates within a single instance; consolidation across eight instances is future work.
- **Proposed vs. measured.** The conductor and stack are specified; the receipt rail is deployed and verified (20/20); the gated full-duplex digital human is proposed.

8. Conclusion

The one-person company is limited not by compute but by authority. The conductor, not the agent, is the product. We specified five duties, identified fan-in as the critical gap, gave a minimal receipt rail, and extended the model vertically to a digital human. The missing layer there is cryptographic identity — which the sovereign layer supplies. The mirror is solved. The right to reflect is not.

References

[1] R. Roy, J. Raiman, S.-g. Lee, T.-D. Ene, R. Kirby, S. Kim, J. Kim, B. Catanzaro. *PersonaPlex: Voice and Role Control for Full Duplex Conversational Speech Models.* arXiv:2602.06053, 2026.

[2] A. Défossez et al. *Moshi: a speech-text foundation model for real-time dialogue.* Kyutai; arXiv:2410.00037, 2024.

[3] NVIDIA. *NVIDIA ACE — Avatar Cloud Engine.* https://github.com/NVIDIA/ACE.

[4] NVIDIA. *Tokkio Digital Human AI Blueprint.* https://github.com/NVIDIA-AI-Blueprints/digital-human.

[5] J. Wang, B. Chen, et al. *Mixture-of-Agents Enhances Large Language Model Capabilities.* arXiv:2406.04692, 2024.